ZentroMart VoiceOps — practice report

Author: Individual project

Repository: https://github.com/Ru0k3/ZentroMart-VoiceOps-capstone-

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LLM used for the recorded full evaluation: `nvidia/nemotron-3.5-lightning-30b-a3b`

System

The implementation follows the four capstone stages: a prompt-only baseline, section-aware RAG over the supplied knowledge base and product catalog, multi-turn order and ticket tools, and Gnani Prisma STT → the support agent → Gnani Timbre TTS. Retrieval uses BM25, retains section IDs and policy metadata, and excludes superseded policies by default. CSV identifiers stay strings. Order details are gated by the dataset verification tool; ticket success is reported only after the tool confirms creation. Voice outputs save the reply, WAV audio, retrieved evidence, and citations, but do not save the recognized customer transcript.

Full practice evaluation

The full saved run contains 52 baseline questions, 52 RAG questions, and 96 conversation turns. The model was `nvidia/nemotron-3.5-lightning-30b-a3b`. Strict pass requires the expected decision, required and forbidden phrase checks, expected tool actions, and all gold evidence IDs where applicable. Phrase matching is literal after normalization, so these are practice-set signals rather than semantic quality scores.

Stage	Cases	Strict pass	Decision pass	Required facts pass	Forbidden facts pass	Gold evidence matched
Prompt-only baseline	52	2 (3.8%)	31/52	10/52	52/52	0/90
RAG	52	8 (15.4%)	39/52	26/52	52/52	30/90 (33.3%)
Conversations	96	2 (2.1%)	54/96	72/96	96/96	2/112 (1.8%)

RAG improved required-fact matches from 10 to 26 and matched 30 gold evidence IDs where baseline matched none. The full conversation run passed 83/96 expected tool-action checks. Its traces showed parallel action proposals, so the agent now executes at most one support action per customer turn and replaces an order-specific answer with a verification request after a failed lookup. A scripted local check confirmed those safeguards. Decision tags and source tags are also recovered when omitted: decisions use a conservative text rule, and citations require substantial overlap with retrieved passage content.

The final conversation run used the updated retrieval and prompt code and completed all 96 turns, with explicit waits after NVIDIA HTTP 429 responses. Its strict score was 2/96; evidence matching was 2/112 (1.8%). These low scores are material limitations: grounded voice success on one targeted warranty case does not imply reliable citation coverage across the full practice suite. The main failure is evidence coverage in conversation mode. I would next test retrieval depth and prompt behavior separately, then make citation selection more reliable from the passages actually supplied to the model and rerun all conversation cases. The low strict-pass counts also need case-by-case review; the current literal phrase checks can miss correct paraphrases, but that does not explain away the evidence-recall result.

Targeted retrieval regression (PQ0002): The practice ASR wording “ele-zero zero one to” previously mapped poorly and unrelated product rows crowded the relevant evidence out. Retrieval now normalizes spoken SKU digits, boosts exact SKU and section-title matches, and excludes other product rows when the query names an exact SKU. The targeted RAG case then retrieved both `products.csv:ELE-0012` and `WAR-2` and matched its required facts. This does not offset the full-set RAG score above. The result identifies a retrieval robustness gap; it does not establish the dataset authors' intent.

Voice run and limits

Gnani Timbre v2.5 with the Kaveri voice generated a valid 8.16-second WAV in a TTS smoke run. A 62.55-second own-voice recording was converted to a 59.59-second, 16 kHz mono WAV; Gnani REST rejected it above its 30-second limit, so the runner falls back to Gnani batch STT for that specific error. STT, RAG, and TTS then completed once and produced a 28.96-second WAV.

The initial voice answer retrieved no `WAR-2` and had no citation. After the Hindi/Hinglish retrieval changes and warranty-intent boost, a later own-voice run retrieved `WAR-2` plus `products.csv:ELE-0012`, answered with the 12-

month Assured warranty and My Orders claim path, and generated a 25.76-second mono 48 kHz WAV. The citations are in the saved evidence metadata and are not read aloud. This verifies one relevant voice case; it does not establish broad multilingual accuracy. Only the builder's own voice and the supplied synthetic dataset were used.

Reproduction

Install dependencies with `pip install -r requirements.txt` and `pip install -e .`; copy `.env.example` to `.env` and supply local keys. The evaluation JSONL files and generated audio are local outputs and are not part of the repository. See `DEMO_PLAN.md` for the short demonstration sequence.